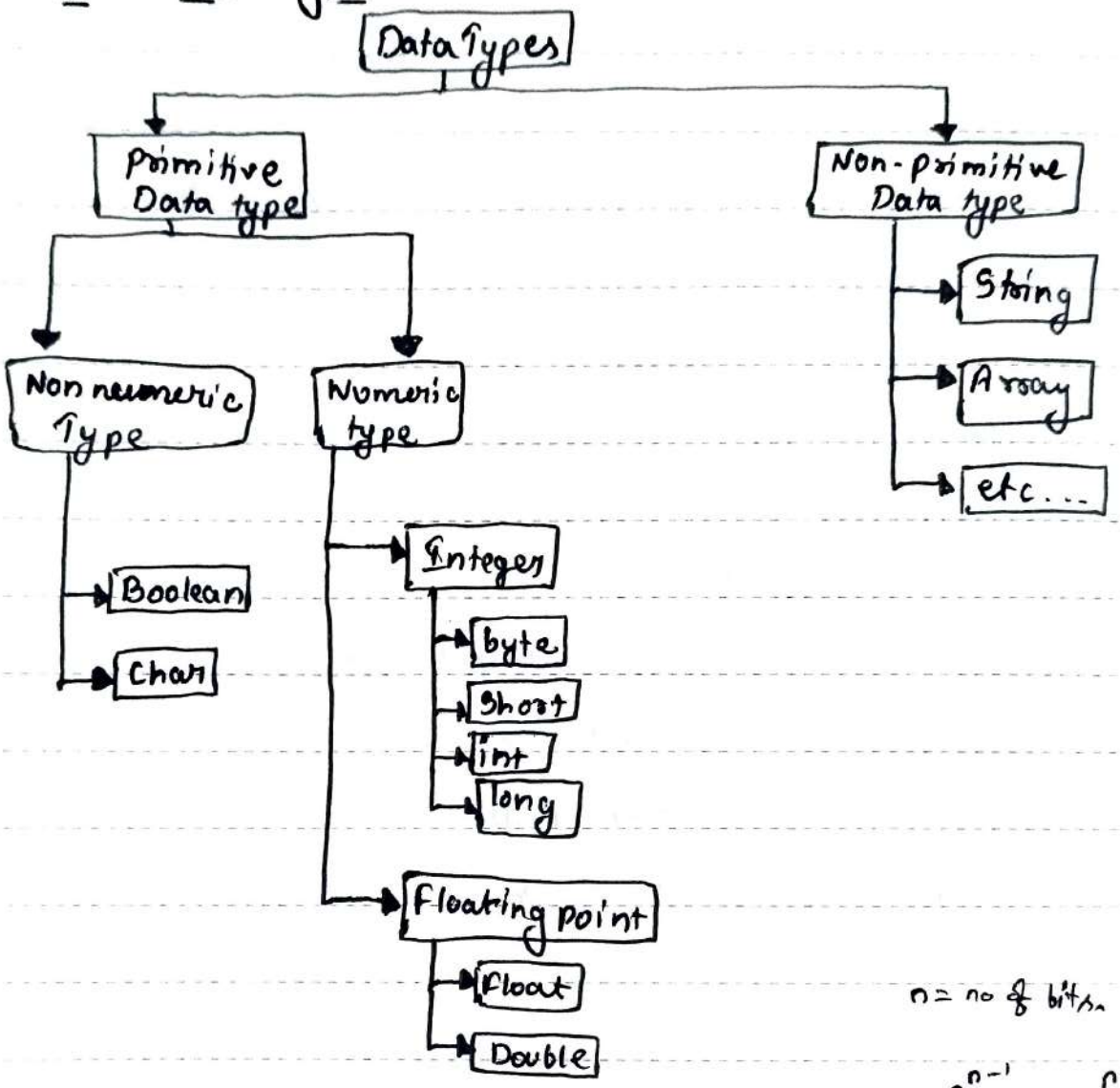
Java Data types :-



n = no of bits.

-2^{n-1} to $2^{n-1}-1$

Type	Desc	Default	Size	Example	range
Boolean	Logical values	false	Not JVM-defined	True, false	–
byte	8-bit signed integer	0	1 byte	10	-128 to 127
char	16-bit ~~signed~~ unicode ~~int~~ character	\u0000	2 byte	'A', '\u0041'	0 to 65,535
Short	16-bit signed integer	0	2 byte	2000	-32,768 to 32,767
int	32-bit signed integer	0	4 byte	1000, -500	-2,147,483,648 to 2,147,483,647
long	64-bit signed integer	0L	8 byte	1234567892L	± 9.22e18

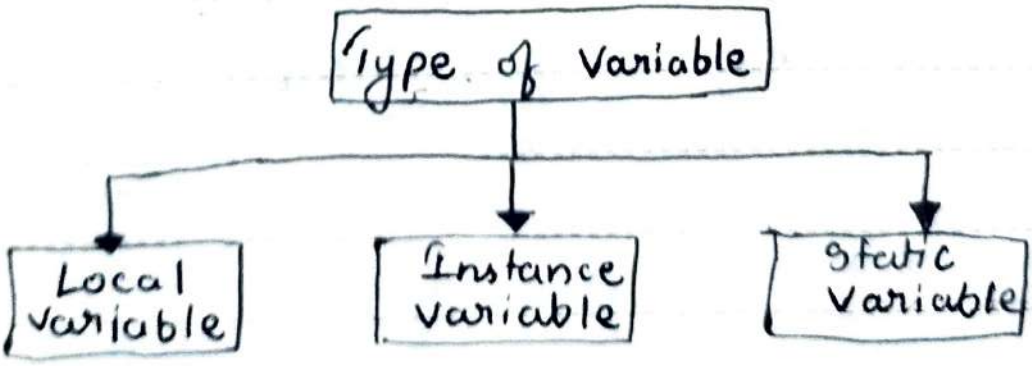
Float	32-bit Floating Point	0.0f	4 bytes	3.14f	~6-7 digits precision
double	64-bit Floating Point	0.0d	8 bytes	3.14159d	~15-16 digits precision

*) In java, any decimal number is treated as "double" by default.

```
double x = 0.5; ✓
double y = 0.5d; ✓
float z = 0.01f; ✓
float s = 0.01; X // compile time error.
```

because: 0.01 -> considered double,
java does not automatically convert double -> float
But, for double, it does not require "d"

Types of Variables in Java:



Local Variable : A variable defined within a block, method or constructor.

*) Local Variable is created at time of declaration & destroyed when function completes its execution.

*) Scope is only within block in which they created.

*) Need to initialize before using it within scope. $\boxed{a = 10;}$

Instance Variable : Known as non-static variables and are declared in a class outside of any method, constructor or block.

*) Created when object is instantiated and destroyed when object is destroyed.

*) Can have access specifiers; "default" access is used if none is specified.

*) Instance variable are not mandatory to initialize, takes default values based on data type (0 for int, null for string, etc...).

*) Scope is throughout class, except in static context.

*) Accessed only through objects of class

*) Instance variables can be initialized using Constructors or instance blocks.

```
class Student {
    public int i; // Declaring without Initialisation

    public Student()
    { // initialising Instance Variable using constructor
      this.i = 10;
    }
}
```

Static Variable : Variables declared with "static" keyword inside a class but outside any method. They are shared among all objects of class and exist for entire lifetime of program.

*) There is only one copy of static variable for entire class, all objects share it.

*) Created at program start, destroyed when program ends.

*) Not mandatory to initialize, default assigned based on data type.

*) Can be accessed via class name, accessing through objects shows compiler warning.

*) Cannot define or be declared inside Instance method locally.

```
class Greeks {
        // Static variable
    static String geek = "Hi welcome!";

    Public static void main (String[] args)
    {
      // Access static without creating object
        sout (Greeks.geek);

      // static int c = 0;
      // Error: static variables cannot be declared
      inside a method.
    }
}
```

Java Operators :-

Arithmetic Operators : +, -, *, %, /
Logical Operators : &&, ||, !
Unary Operators : --, ++
Ternary Operators : cond ? true : false;
Assignment Operator : +=, -=, =, /=, %=, *=
Bitwise & Shift Operator : &, |, ^, <<, >>, >>>